Further Regression Methods

Linear Mixed Models (LMMs)

In Richard McElreath's book "Statistical Rethinking", chapters 13 and 14 deal with multilevel models. In Gelman's Book, see " Part 2A: Multilevel Models" and the following.

One very graspable introduction to LMMs is this video.

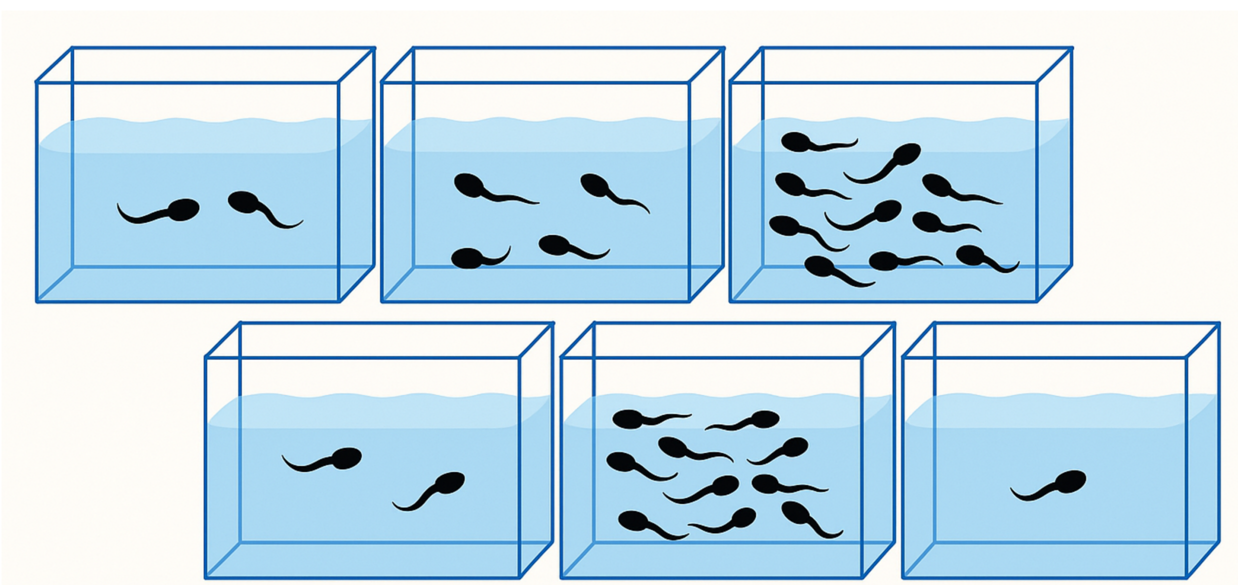
We have already come across a Linear Mixed Model (LMM) in the first chapter, where we used the `rethinking` and `lme4` packages to fit a model with random intercepts to determine our Intraclass Correlation Coefficients (ICCs). They are called *random* intercept since they are drawn from a common normal distribution, of which the parameters (mean and standard deviation) are estimated from the data.

The **main reason** for using LMMs is that they allow us to model data with hierarchical structure, such as repeated measures (of the same subject) or nested data (math scores in different schools within different school districts).

- For example, in our Peter and Mary data set where they measured ROMs, we had two measurements per subject. The two measurements are *clustered* within the subject. They are not independent. If Mary measures a very high ROM, changes are that Peter as well measures a high value. Hence, the classical linear regression model (which assumes the rows/obervations in our data set are independent) is not appropriate and would lead to smaller standard errors.
- Another example is our *RESOLVE* trial. We are conducting a cluster randomized trial (CRT) with 20 clusters (physiotherapy offices) and over 200 patients. Measurements within a cluster could be correlated (more similar compared to other clusters). One could also think of the physiotherapists performing the treatments as clusters, since - potentially - the outcomes for all their patients could be correlated due to a more similiar treatment from the same therapists compared to others. Here the outcome measures are nested within the physiotherapists which are nested within the physiotherapy offices (assuming they only work for one office).

Example: Reed Frog tadpole mortality - two levels

Let's start with Example 13.1 from McElreath's book (p.401) (Tadpoles in tanks, image by GPT4o).



```r
library(rethinking)
```

```
## Loading required package: cmdstanr

## This is cmdstanr version 0.8.1.9000

## - CmdStanR documentation and vignettes: mc-stan.org/cmdstanr

## - CmdStan path: /Users/juergen/.cmdstan/cmdstan-2.36.0

## - CmdStan version: 2.36.0

## Loading required package: posterior

## This is posterior version 1.6.1

##
## Attaching package: 'posterior'

## The following objects are masked from 'package:stats':
##
##     mad, sd, var

## The following objects are masked from 'package:base':
##
##     %in%, match

## Loading required package: parallel

## rethinking (Version 2.42)

##
## Attaching package: 'rethinking'

## The following object is masked from 'package:stats':
##
##     rstudent
```

```r
library(lme4)
```

```
## Loading required package: Matrix
```

```r
data(reedfrogs)
d <- reedfrogs
str(d)
```

```
## 'data.frame':    48 obs. of  5 variables:
##  $ density : int  10 10 10 10 10 10 10 10 10 10 ...
##  $ pred    : Factor w/ 2 levels "no","pred": 1 1 1 1 1 1 1 1 2 2 ...
##  $ size    : Factor w/ 2 levels "big","small": 1 1 1 1 2 2 2 2 1 1 ...
##  $ surv    : int  9 10 7 10 9 9 10 9 4 9 ...
##  $ propsurv: num  0.9 1 0.7 1 0.9 0.9 1 0.9 0.4 0.9 ...
```

Variables are explained here (see also p.401 in the book). The original paper can be found here.

The number of surviving tadpoles *surv* should be modeled. Now, each row is a different "tank" (48 in total), an experimental environment that contains tadpoles. Obviously, the tanks are one option for *clustering* the data, assuming the tadpoles within the same tank live in a more similar environment than those in different tanks (or get a more similiar "treatment"). We could rewrite the data set to have one row per individual tadpole (instead of number of survived a binary variable) and the cluster variable called *tank* (exercise 1). We would then model the survival probability of tadpoles (1 could mean *survived* and 0 could mean *died*). Formally, this would be a so-called logistic regression (see below), since our outcome is binary (*survided/died*).

When estimating the number of surviving tadpoles we could consider the following options:

- **Complete pooling** (exercise 6): Estimate the overall survival probability (ignoring the tank effect). In this case, one might miss between-variance from the different tanks. A single intercept is used for all tanks.
- **No pooling**: Estimate the survival probability for each tank (ignoring the overall effect completely). Here, the tanks do not "know" anything from one another. Information of one tank is not used for the other tanks. An intercept is estimated for each tank (48 in total). This approach would be equivalent to using indicator variables for each tank.
- **Partial pooling**: Estimate the survival probability for each tank, but use information from the other tanks.

Model 13.1. Partial pooling using regularizing priors Let's start with the second model (intercept for each tank) with a regularizing prior. We want to model the number of surviving tadpoles in the tanks by using a binomial distribution with different survival probabilities for each tank.

Here is our model in the typical notation:

$$\begin{aligned} S_i &\sim Binomial(N_i, p_i) \\ logit(p) &= \alpha_{TANK}[i] \\ \alpha_j &\sim \text{Normal}(0, 1.5) \text{ for } j = 1, \ldots, 48 \end{aligned}$$

This is a **varying intercept model** (the simplest kind of **varying effects** model) with 48 intercepts (one for each tank).

The number of surviving tadpoles S_i in tank i is modeled as a binomial distribution with N_i trials (the number of tadpoles in the tank) and p_i the survival probability of tank i. The **regularizing priors** for α_j is a normal distribution with mean 0 and standard deviation 1.5.

Technically, **we could use different priors for each intercept**, but this would be a bit cumbersome and we do not have useful prior information about the survival probabilities of the tanks.

Notice that we use a `logit` link function to transform the probability p (range $(0, 1)$) to the range $(-\infty, \infty)$. This way, we can use a normal distribution for the prior of the intercepts. The inverse of the logit function is also available in `rethinking`: `inv_logit`. `rethinking::inv_logit(0)`gives the probability of 0.5. So, we assume a coin flip for the survival probability of the tadpoles in the tanks *a priori* (maybe not the worst starting point if you have no clue).

What survaival probabilities do we assume a priori? Well, according to the normal distribution with a standard deviation of 1.5 we assume that the 95% of the logit-transformed survival probabilities are between -3 and 3. This means that the survival probabilities in the tanks are in:

```
c(rethinking::inv_logit(-3), rethinking::inv_logit(3))
```

```
## [1] 0.04742587 0.95257413
```

This assumption seems rather reasonable and survival probabilities are not constrained very much.

Next we define the data set for the model and use the `ulam` function from the `rethinking` package to fit the model.

```
library(rethinking)

d$tank <- 1:nrow(d)
dat <- list(S = d$surv,
            N = d$density,
            tank = d$tank)

# approximate posterior
set.seed(122)
m13.1 <- ulam(
  alist(
    S ~ dbinom(N, p),
    logit(p) <- a[tank],
    a[tank] ~ dnorm(0, 1.5)
  ) , data = dat,
  chains = 4,
  log_lik = TRUE,
  cores = detectCores() - 1)
```

```
## Running MCMC with 4 chains, at most 7 in parallel, with 1 thread(s) per chain...
##
## Chain 1 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 1 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 1 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 1 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 1 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 1 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 1 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 1 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 1 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 1 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 1 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 1 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 2 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 2 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 2 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 2 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 2 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 2 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 2 Iteration: 501 / 1000 [ 50%]  (Sampling)
```

```
## Chain 2 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 2 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 2 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 2 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 2 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 3 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 3 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 3 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 3 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 3 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 3 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 3 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 3 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 3 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 3 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 3 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 3 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 4 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 4 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 4 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 4 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 4 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 4 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 4 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 4 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 4 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 4 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 4 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 4 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 1 finished in 0.1 seconds.
## Chain 2 finished in 0.1 seconds.
## Chain 3 finished in 0.1 seconds.
## Chain 4 finished in 0.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.1 seconds.
## Total execution time: 0.2 seconds.
```

```r
precis(m13.1, depth = 2)
```

```
##               mean        sd        5.5%       94.5%      rhat ess_bulk
## a[1]   1.717154006 0.7747045  0.54616751  3.00284170 1.0000687 3842.501
## a[2]   2.428617822 0.9211741  1.06824055  4.00015385 1.0012690 3260.212
## a[3]   0.752957586 0.6606081 -0.30096467  1.87725390 0.9994289 3657.926
## a[4]   2.392671596 0.9029471  1.04232310  3.96444320 0.9999341 4901.360
## a[5]   1.698600986 0.7501296  0.56599784  2.95760400 1.0010747 3583.085
## a[6]   1.709016626 0.7333373  0.61628599  2.93631575 1.0043948 3791.625
## a[7]   2.394662704 0.8967971  1.05357265  3.91037710 1.0021691 4633.411
## a[8]   1.751520527 0.8358778  0.51949731  3.11502600 1.0010314 2908.886
## a[9]  -0.368916633 0.6263820 -1.40362045  0.63108741 1.0014184 5011.742
## a[10]  1.721989971 0.7876527  0.53971156  3.01818445 1.0064067 4743.990
## a[11]  0.762625202 0.6504815 -0.23379561  1.84594325 1.0042895 4356.338
## a[12]  0.360839662 0.6212530 -0.61653294  1.35696620 1.0003393 5427.338
## a[13]  0.747187659 0.6556836 -0.26629876  1.86578240 1.0047650 4965.966
```

```
## a[14]  0.004006226 0.6030814 -0.96596033  0.95626616 1.0015451 3860.382
## a[15]  1.698071502 0.7557423  0.58532612  3.00604665 1.0044770 4060.982
## a[16]  1.718281276 0.7589570  0.55788234  3.01417210 0.9995458 4085.653
## a[17]  2.540229640 0.6715384  1.51315760  3.67929090 1.0045229 3551.210
## a[18]  2.143608256 0.6099572  1.24664285  3.19176460 1.0027772 4601.028
## a[19]  1.814013282 0.5661173  0.96992496  2.74369310 1.0008102 3875.396
## a[20]  3.123650833 0.8415721  1.90725255  4.55287270 1.0001108 4325.296
## a[21]  2.144628056 0.6206063  1.21321885  3.18875240 1.0003712 4533.942
## a[22]  2.147574902 0.5862451  1.27518855  3.13301315 1.0054417 3115.241
## a[23]  2.158113205 0.6232680  1.23542735  3.20262840 1.0031826 4470.846
## a[24]  1.559385117 0.5054760  0.81471935  2.39450025 1.0050444 4677.140
## a[25] -1.100464102 0.4645629 -1.90970175 -0.37079074 1.0031187 4404.834
## a[26]  0.066658204 0.3997793 -0.57122361  0.71319580 1.0064293 5082.203
## a[27] -1.543870059 0.4969425 -2.35267600 -0.80927273 1.0049318 4888.677
## a[28] -0.545686639 0.3798291 -1.16412750  0.05957414 1.0141192 4539.855
## a[29]  0.076432999 0.3891808 -0.54687735  0.68588656 1.0063621 4372.045
## a[30]  1.317149701 0.4699257  0.61627877  2.10864310 1.0045814 4237.156
## a[31] -0.730232329 0.4142389 -1.41215210 -0.08699833 1.0056791 4674.458
## a[32] -0.380477602 0.3973004 -1.02838145  0.22966973 1.0027171 4757.953
## a[33]  2.854476850 0.6688006  1.86231770  3.93923980 1.0028748 4265.441
## a[34]  2.465178502 0.6016136  1.57307340  3.49223990 1.0034665 4026.537
## a[35]  2.475988085 0.6001981  1.62509180  3.45605210 1.0022452 3520.649
## a[36]  1.906146592 0.4793998  1.17540695  2.69468780 1.0091406 3637.720
## a[37]  1.903268812 0.4848702  1.17045730  2.68925485 0.9999314 4580.692
## a[38]  3.377291100 0.7728682  2.21480150  4.70722490 0.9999677 3559.742
## a[39]  2.448119286 0.5794139  1.58287005  3.42906255 1.0005167 4544.304
## a[40]  2.154826607 0.5110004  1.38358250  3.05123950 1.0057481 3928.734
## a[41] -1.924960111 0.4805039 -2.72391445 -1.20680920 1.0003669 3217.189
## a[42] -0.639489130 0.3594361 -1.20374385 -0.07755664 1.0036102 3891.702
## a[43] -0.515434665 0.3438150 -1.06684055  0.01709138 1.0067069 3206.309
## a[44] -0.399514162 0.3554107 -0.97334898  0.16487656 1.0043647 5212.318
## a[45]  0.519393681 0.3592609 -0.05772033  1.08264245 1.0022088 4807.785
## a[46] -0.636130179 0.3468987 -1.20742535 -0.10853365 1.0071298 3963.335
## a[47]  1.919195782 0.4916630  1.16999095  2.74101715 1.0000987 3428.411
## a[48] -0.061061407 0.3286406 -0.57313044  0.45005764 1.0009603 3435.612
```

```r
# expected survival in tank 1
rethinking::logistic(1.717154006) # same as inv_logit(1.717154006)
```

```
## [1] 0.8477619
```

The R-output from `precis(m13.1, depth = 2)`shows the posterior means and standard deviations of our 48 intercepts (48 tanks with tadpoles). These intercepts are on the scale $(-\infty, \infty)$.

Note that you can always switch from the range $(-\infty, \infty)$ to a probability using the `logistic` function from the `rethinking` package - these functions are called *squashing functions* for a reason. The `inv_logit`function does the same.

$$\text{logistic}(x) = \text{inv\_logit}(x) = \frac{1}{1 + e^{-x}}$$

The other way around works as well, using the `logit` function.

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

Let's quickly see if these are indeed inverse functions of each other -> exercise 5.

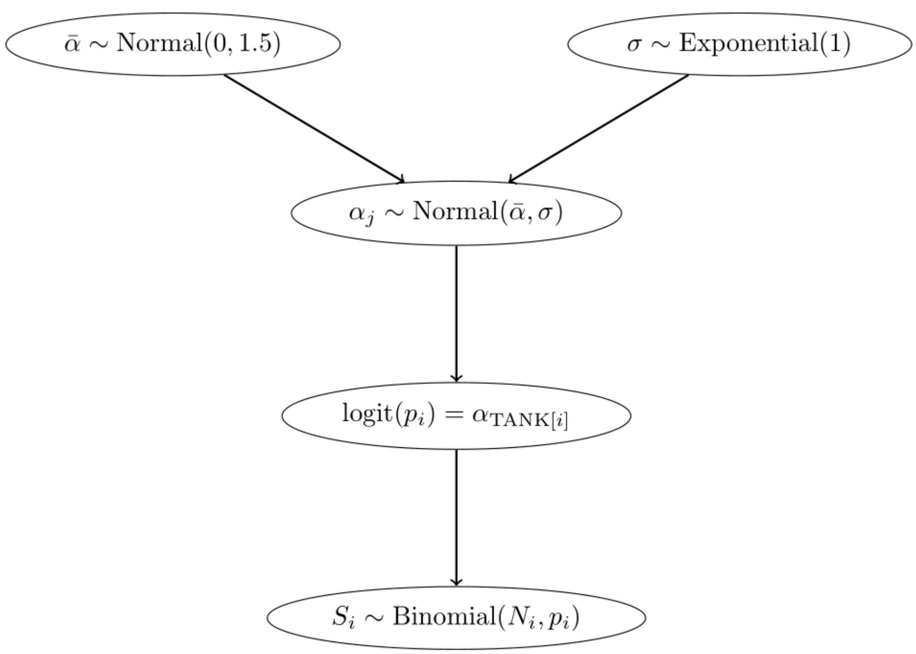
Model 13.2: Partial pooling using adaptive pooling **Next**, we use **adaptive pooling**. For this, we use priors for the `a[tank]` parameters, but know they *know* from each other via the prior for $\bar{\alpha}$. This model learns from the data how much the intercepts should be pulled towards a common mean. Now we have an additional level in the model hierarchy:

- In the **top level**, the outcome is S, the parameters are the intercepts $\alpha_{TANK[i]}$, and the priors are $\alpha_j \sim N(\bar{\alpha}, \sigma)$.
- In the **second level**, the "outcome" are the intercepts $\alpha_{TANK[i]}$, the parameters are $\bar{\alpha}$ and σ, and the priors are $\bar{\alpha} \sim N(0, 1.5)$ and $\sigma \sim \text{Exponential}(1)$.

$$
\begin{aligned}
S_i &\sim \text{Binomial}(N_i, p_i) \\
\text{logit}(p) &= \alpha_{\text{TANK}[i]} \\
\alpha_j &\sim \text{Normal}(\bar{\alpha}, \sigma) \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

Since the parameters in the prior of the α_j have also priors (last two lines are new), we have a *multilevel* model. In this case, we have *two* levels and the parameters $\bar{\alpha}$ and σ are called **hyperparameters**. The priors of the hyperparameters are called **hyperpriors**. The **amount or regularization is controlled by the hyperparameter** σ and is learned from the data. σ controls, how much the intercepts are allowed to deviate from the mean $\bar{\alpha}$. The smaller σ, the more the intercepts are pulled towards the mean $\bar{\alpha}$.

Below, we see the model hierarchy:



We now fit the model using the `ulam` function from the `rethinking` package.

```
library(conflicted)
conflicts_prefer(posterior::var)
```

```
## [conflicted] Will prefer posterior::var over any other package.
```

```r
conflicts_prefer(posterior::sd)
```

```
## [conflicted] Will prefer posterior::sd over any other package.
```

```r
m13.2 <- ulam(
  alist(
    S ~ dbinom(N, p),
    logit(p) <- a[tank],
    a[tank] ~ dnorm(a_bar, sigma),
    a_bar ~ dnorm(0, 1.5),
    sigma ~ dexp(1)
  ) , data = dat,
  chains = 4,
  log_lik = TRUE,
  cores = detectCores() - 1)
```

```
## Running MCMC with 4 chains, at most 7 in parallel, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 1000 [  0%]  (Warmup)
## Chain 1 Iteration:  100 / 1000 [ 10%]  (Warmup)
## Chain 1 Iteration:  200 / 1000 [ 20%]  (Warmup)
## Chain 1 Iteration:  300 / 1000 [ 30%]  (Warmup)
## Chain 1 Iteration:  400 / 1000 [ 40%]  (Warmup)
## Chain 1 Iteration:  500 / 1000 [ 50%]  (Warmup)
## Chain 1 Iteration:  501 / 1000 [ 50%]  (Sampling)
## Chain 1 Iteration:  600 / 1000 [ 60%]  (Sampling)
## Chain 1 Iteration:  700 / 1000 [ 70%]  (Sampling)
## Chain 1 Iteration:  800 / 1000 [ 80%]  (Sampling)
## Chain 1 Iteration:  900 / 1000 [ 90%]  (Sampling)
## Chain 1 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 2 Iteration:    1 / 1000 [  0%]  (Warmup)
## Chain 2 Iteration:  100 / 1000 [ 10%]  (Warmup)
## Chain 2 Iteration:  200 / 1000 [ 20%]  (Warmup)
## Chain 2 Iteration:  300 / 1000 [ 30%]  (Warmup)
## Chain 2 Iteration:  400 / 1000 [ 40%]  (Warmup)
## Chain 2 Iteration:  500 / 1000 [ 50%]  (Warmup)
## Chain 2 Iteration:  501 / 1000 [ 50%]  (Sampling)
## Chain 2 Iteration:  600 / 1000 [ 60%]  (Sampling)
## Chain 2 Iteration:  700 / 1000 [ 70%]  (Sampling)
## Chain 2 Iteration:  800 / 1000 [ 80%]  (Sampling)
## Chain 2 Iteration:  900 / 1000 [ 90%]  (Sampling)
## Chain 2 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 3 Iteration:    1 / 1000 [  0%]  (Warmup)
## Chain 3 Iteration:  100 / 1000 [ 10%]  (Warmup)
## Chain 3 Iteration:  200 / 1000 [ 20%]  (Warmup)
## Chain 3 Iteration:  300 / 1000 [ 30%]  (Warmup)
## Chain 3 Iteration:  400 / 1000 [ 40%]  (Warmup)
## Chain 3 Iteration:  500 / 1000 [ 50%]  (Warmup)
## Chain 3 Iteration:  501 / 1000 [ 50%]  (Sampling)
## Chain 3 Iteration:  600 / 1000 [ 60%]  (Sampling)
## Chain 3 Iteration:  700 / 1000 [ 70%]  (Sampling)
## Chain 3 Iteration:  800 / 1000 [ 80%]  (Sampling)
## Chain 3 Iteration:  900 / 1000 [ 90%]  (Sampling)
```

```
## Chain 3 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 4 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 4 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 4 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 4 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 4 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 4 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 4 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 4 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 4 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 4 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 4 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 4 Iteration: 1000 / 1000 [100%]  (Sampling)
```

```
## Chain 4 Informational Message: The current Metropolis proposal is about to be rejected because of the
## Chain 4 Exception: normal_lpdf: Scale parameter is 0, but must be positive! (in '/var/folders/pm/jd6n
## Chain 4 If this warning occurs sporadically, such as for highly constrained variable types like covar
## Chain 4 but if this warning occurs often then your model may be either severely ill-conditioned or mi
## Chain 4
```

```
## Chain 1 finished in 0.1 seconds.
## Chain 2 finished in 0.1 seconds.
## Chain 3 finished in 0.1 seconds.
## Chain 4 finished in 0.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.1 seconds.
## Total execution time: 0.2 seconds.
```

```r
precis(m13.2, depth = 2)
```

```
##               mean        sd        5.5%       94.5%      rhat ess_bulk
## a[1]   2.1360104954 0.8661191  0.83424198  3.553682700 1.0026805 3270.394
## a[2]   3.0781335085 1.1236851  1.43096300  4.984337950 1.0021514 3333.772
## a[3]   1.0172166998 0.6728834  0.02487391  2.122557100 1.0017997 4433.035
## a[4]   3.0539084560 1.0673010  1.45577985  4.862345550 0.9991935 3292.307
## a[5]   2.1460704153 0.8885053  0.82128835  3.632871100 1.0012996 3243.212
## a[6]   2.1295681940 0.8491091  0.88580128  3.547453050 1.0068927 4014.260
## a[7]   3.0995747675 1.1423048  1.49656170  5.081114500 1.0000324 2876.988
## a[8]   2.1399446424 0.9003458  0.81845163  3.682066500 1.0014370 3989.652
## a[9]  -0.1714730123 0.6191399 -1.18887420  0.817493905 1.0011313 3485.167
## a[10]  2.0975422705 0.8315647  0.86475233  3.491159800 1.0003507 3475.655
## a[11]  0.9972846111 0.6886670 -0.06780398  2.108900800 1.0035689 6346.111
## a[12]  0.5808274429 0.6321307 -0.39005998  1.589965800 1.0073548 4009.231
## a[13]  1.0044482950 0.6869233 -0.06659326  2.145820400 1.0062364 5228.172
## a[14]  0.1943572393 0.5958137 -0.75088828  1.137035850 1.0090229 4741.338
## a[15]  2.1509188614 0.8679934  0.90383891  3.653387850 1.0031940 3312.879
## a[16]  2.1333640552 0.9017144  0.80812878  3.653567300 1.0126010 4233.256
## a[17]  2.9192000930 0.8415071  1.68621300  4.328422500 1.0040634 4410.077
## a[18]  2.3925664980 0.6551941  1.42167145  3.513079150 1.0056149 4144.799
## a[19]  2.0077513930 0.5900888  1.11100110  2.993477400 1.0059627 3633.662
## a[20]  3.6725677750 1.0278681  2.20375370  5.475839300 1.0049323 3236.751
## a[21]  2.4059350610 0.6759614  1.38493480  3.559096600 1.0015502 5132.563
```

```
## a[22]  2.4065305525 0.6796675  1.42094305  3.541692650 0.9991290 3831.139
## a[23]  2.3861532095 0.6514720  1.42360935  3.472103550 1.0043729 4128.680
## a[24]  1.7089185740 0.5393019  0.87925440  2.627249950 1.0048578 4608.699
## a[25] -1.0120972883 0.4472418 -1.73198540 -0.326257340 1.0014661 4734.657
## a[26]  0.1608543317 0.3886832 -0.46547174  0.767354565 1.0023246 4941.014
## a[27] -1.4196695957 0.4932724 -2.25709745 -0.685836175 1.0028562 4123.586
## a[28] -0.4766474294 0.4060361 -1.14774660  0.151370735 1.0052220 3931.755
## a[29]  0.1694769825 0.4170077 -0.49255198  0.859140040 1.0020527 4739.882
## a[30]  1.4511140548 0.5133738  0.66183801  2.306309700 1.0011993 4978.641
## a[31] -0.6385969972 0.4191422 -1.30238420  0.002327188 1.0021641 4600.143
## a[32] -0.3101703440 0.3728743 -0.90309567  0.287795040 0.9998292 5387.376
## a[33]  3.1774488200 0.7655335  2.08349505  4.549344000 1.0049724 3861.331
## a[34]  2.7053424900 0.6559522  1.73698945  3.801545200 1.0012059 4418.032
## a[35]  2.7221132200 0.6273888  1.81337830  3.780182650 1.0009124 3645.004
## a[36]  2.0546272795 0.4965714  1.31339900  2.878739800 1.0062188 4941.400
## a[37]  2.0584841305 0.5188968  1.26758275  2.917035500 1.0015659 4474.923
## a[38]  3.8972652500 0.9072140  2.63982515  5.508591850 1.0012459 3004.701
## a[39]  2.6974989445 0.6268090  1.74492005  3.751004950 0.9991983 3801.943
## a[40]  2.3462939655 0.5804959  1.47061030  3.318163350 1.0087612 3756.208
## a[41] -1.8135897265 0.4813845 -2.60774680 -1.085182300 1.0000863 3538.261
## a[42] -0.5797062286 0.3540843 -1.15873190 -0.026373790 1.0025961 3932.502
## a[43] -0.4553320681 0.3423639 -1.00197960  0.080289540 1.0064972 4788.740
## a[44] -0.3294683952 0.3311178 -0.86400261  0.191963110 1.0099484 4098.959
## a[45]  0.5781168532 0.3537311  0.02023151  1.152294850 1.0051871 4534.473
## a[46] -0.5734202532 0.3340746 -1.11167870 -0.052582168 1.0045535 3855.996
## a[47]  2.0729423255 0.5180042  1.30783735  2.935491000 1.0033394 3591.954
## a[48] -0.0003544386 0.3378852 -0.52970810  0.533022400 1.0060391 4423.905
## a_bar  1.3420237045 0.2527708  0.94567161  1.760416550 1.0011880 2994.562
## sigma  1.6149891850 0.2052025  1.30750950  1.961327050 0.9998544 1674.452
```

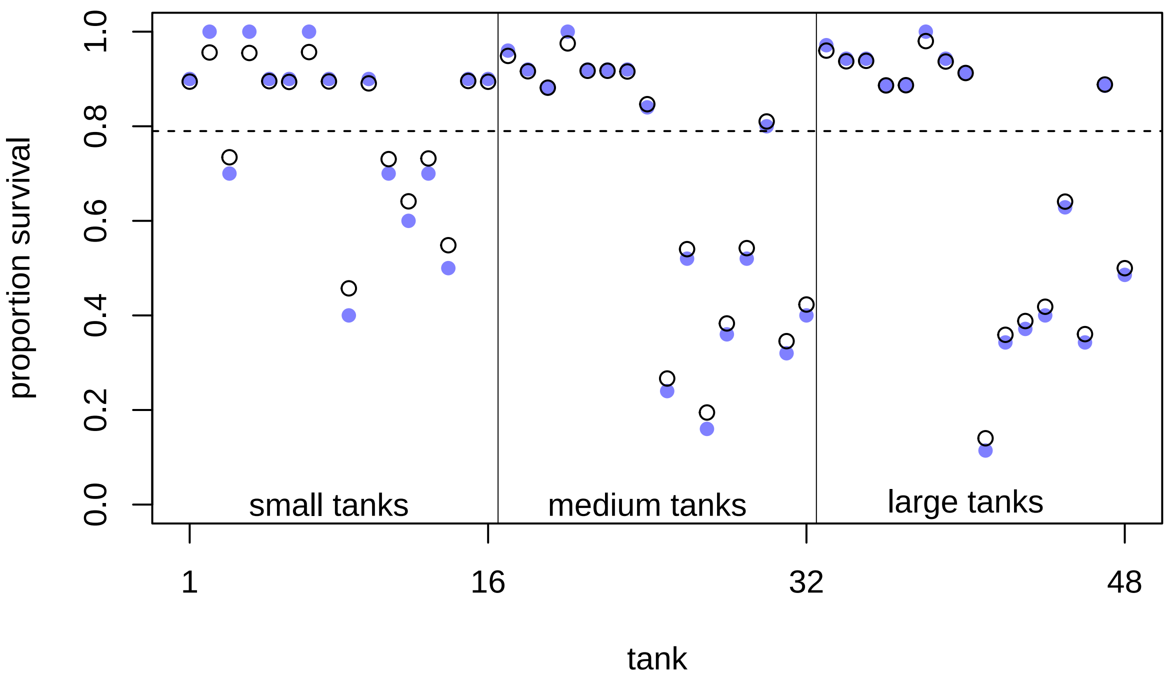
Now, the output of `precis(m13.2, depth = 2)` additionally shows the posterior means and standard deviations of the hyperparameters $\bar{\alpha}$ (`a_bar`) and σ (`sigma`).

Conceptual difference between model 13.1 and 13.2 **Model 13.1** is not a multilevel model, we just tell the model there are 48 tanks and each tank has its own intercept. For each intercept, we use the same prior distribution ($Normal(0, 1.5)$), but we could use different ones if we had knowledge about the survival probabilities of some of the tanks. Note that the priors themselves are not connected via another level. They do their thing independently.

- By *decreasing* the variance of these priors, we are dampening the influence of the data on the intercepts which are drawn towards to overall mean.
- *Increasing* the variance of the priors to a value large enough will result in the raw survival probabilities for each tank. This would be the same as if we took indicator variables for each tank. See exercise 2.

Model 13.2 is a **multilevel model**. Now the model *learns from the data* from which overall mean and variance the intercepts are drawn ($Normal(\bar{\alpha}, \sigma)$). The prior for $\bar{\alpha}$ expresses our belief about the mean survival probability, while the prior for σ expresses our belief about the variability of the survival probabilities.

Figure 13.1 from the book depicts the pooling effect of model 13.2. with respect to tadpole tank size.



In exercise 2 we make the difference between the models 13.1 and 2 clearer. Figure 13.1 from the book shows the posterior survival probabilities for each tank from model 13.2 (black circles) and raw survival probabilities (blue filled points). What we can see is that the estimated survival probabilities for the hierarchical model are drawn towards the mean overall survival probability across all tanks (dashed line; $logistic(\bar{\alpha}) \approx 0.79$).

- In large tanks, there is less shrinkage (enough information/observations), while in small tanks the estimates are pulled more towards the mean (information is used from other tanks).
- The more extreme the raw survival probability (with respect to the overall survival probability), the more it is pulled towards the mean.

This is the **pooling** effect of the model. The phenomenon is called **shrinkage**.

See also exercise 3 to improve your understanding of two models.

Compare models 13.1 and 13.2 with respect to predictive ability on unseen data In QM2 we talked about the difference between explanatory vs. predictive models. If you are interested, you can read section 7.4 (p.217) in the book.

In a **prediction context** we ask ourselves: How well do the models *predict* the data (which was measured using residuals $y_i - \hat{y}_i$ on the *training set* in regression)?

There are many nuances but the **main paradigm** (also in classical machine learning) is to *fit* the model on a **training set** and *evaluate* the model on a **test set**. Should you be interested in the (admittedly very technical) details, I can recommend the book *Elements of Statistical Learning* by Hastie, Tibshirani and Friedman (free here).

During **training**, the model tries to learn the underlying structure of the data.

If the model is too complex, the model can *overfit*, which means that the model learns too much of the noise in the data instead of the underlying structure.

If the model is too simple, the model can *underfit* the training data, which means that the model learns the underlying structure of the data too vaguely.

The test set is used to evaluate the model's performance on unseen data. This is the ultimate and brutally honest test of the model's predictive ability.

For instance, if you think back to our body height prediction example in QM2, we could fit the model on 80% of people (training set) and see how precise the model is on the remaining 20% of people (test set). Let's do this in exercise 7.

Remember, in prediction we do not care about interpretability of the model. We could have a blackbox model (e.g. neural networks) and still be happy with it. This performance can often be surprisingly good (like in the case of hand digit recognition reaching error rates as low as 0.09%), or surprisingly bad (when trying to predict some binary outcome using logistic regression in applied health research where the performance is often not far away from the base rate probability).

In order to estimate the predictive ability (on an unseen test set) of a model, there are many criteria available. One of them is the **WAIC**:

```
WAIC(m13.1)
```

```
##       WAIC      lppd  penalty  std_err
## 1 216.2614 -81.72813 26.40257 4.679153
```

```
WAIC(m13.2)
```

```
##       WAIC      lppd  penalty  std_err
## 1 200.3715 -79.18461 21.00113 7.205089
```

```
compare(m13.1, m13.2, func = WAIC) # change func to other criteria if needed.
```

```
##           WAIC       SE    dWAIC      dSE    pWAIC       weight
## m13.2 200.3715 7.205089  0.00000       NA 21.00113 0.9996456832
## m13.1 216.2614 4.679153 15.88993 3.798405 26.40257 0.0003543168
```

WAIC stands for *Widely Applicable Information Criterion* and is a measure of prediction accuracy on new data (out of sample deviance); see p.219 and p.404 in Statistical Rethinking. As it is pointed out in the book "Using tools like PSIS and WAIC is much easier than understanding them."

We do not go into the details of WAIC, but as a **first dirty rule of thumb**, the model with the lower WAIC is preferred. One should never solely rely on this criterion, especially not if one is interested in causal questions.

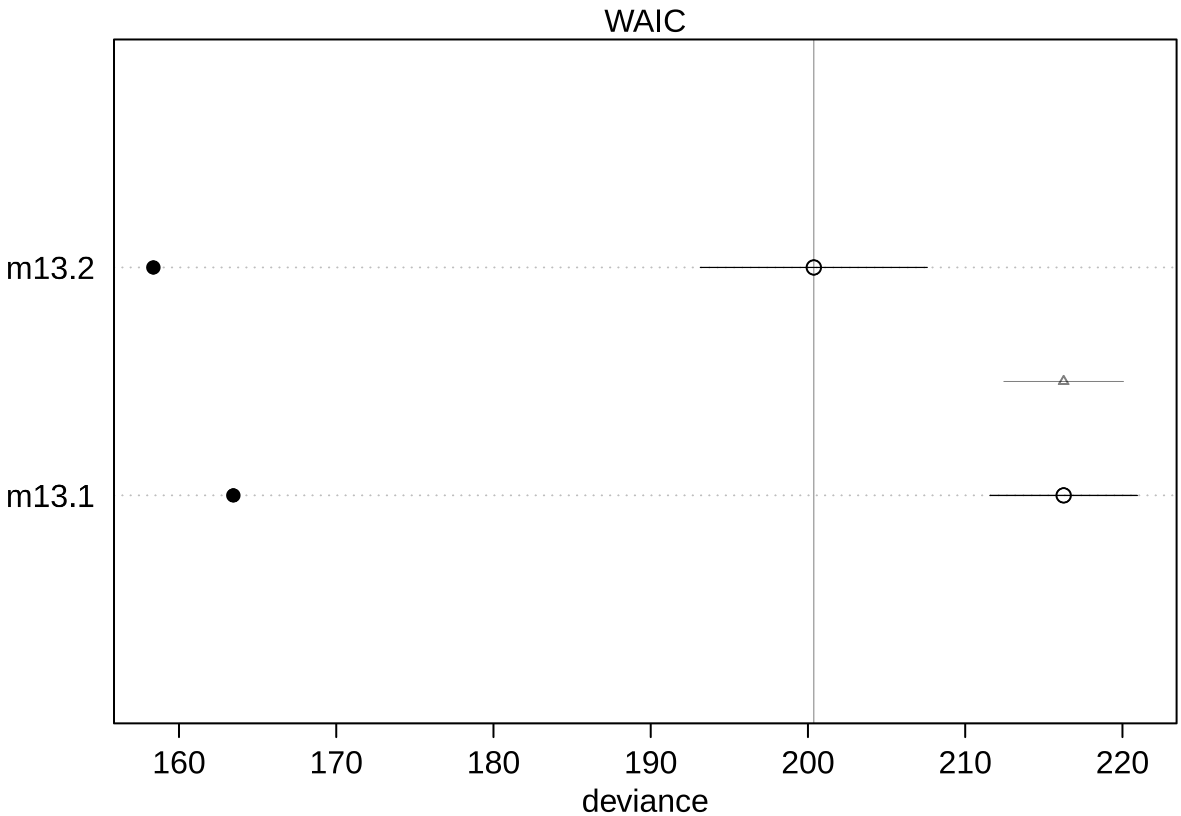
What are the columns?

- `WAIC`: guess of the out-of-sample deviance.
- `SE`: standard error of the WAIC estimate. This is a measure of uncertainty of the WAIC.
- `dWAIC`: difference of each WAIC from the best model (with the *smallest* WAIC).
- `dSE`: standard error of this difference. Here, the standard error of the difference (3.79) is much smaller compared to the difference in WAIC between the models (15.88993). **Hence, one can distinguish between the models using WAIC. → m13.2 is preferred** (with respect to predictive ability on unseen data).
- `pWAIC`: prediction penalty. This is close to the number of parameters in the posterior.
- `weight`: the Akaike weight, a measure of relative support for each model. In our case, it is almost 1 for model 13.2 and almost 0 for model 13.1.

Feel free to read p.227 in the book, specifically before you use WAIC in a publication or thesis.

One can also plot this information for a better overview:

```r
plot(compare(m13.1, m13.2, func = WAIC))
```



The filled points show how the model performs on the data it was trained on. Typically, this is lower compared to the out-of-sample deviance (open points) for both models. Section 13.2 (especially Figure 13.3.) in the book shows that partial pooling also yields (on average) better predictions compared to no pooling, it's a tradeoff between over- and underfitting.

Frequentist version model 13.2 In the Frequentist setting we can use the `lme4` package to fit the analog to model 13.2. To be more specific, we use a generalized linear mixed model (GLMM). *Generalized* because we assume a binomial distribution for the response variable, while usually we have a continuous outcome variable (like body height in cm).

Let's fit the model and explain the outcome first. S is the number of surviving tadpoles, N is the number of tadpoles in the tank.

```r
library(lme4)

m_lmer <- glmer(
  cbind(S, N - S) ~ (1 | tank),
  data = dat,
  family = binomial(link = "logit"),
  control = glmerControl(optimizer = "bobyqa")
)
summary(m_lmer)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
```

```
##  Family: binomial  ( logit )
## Formula: cbind(S, N - S) ~ (1 | tank)
##    Data: dat
## Control: glmerControl(optimizer = "bobyqa")
##
##      AIC      BIC   logLik -2*log(L)  df.resid
##    271.6    275.3   -133.8     267.6        46
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.5981 -0.2255  0.1267  0.2457  0.9677
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  tank   (Intercept) 2.459    1.568
## Number of obs: 48, groups:  tank, 48
##
## Fixed effects:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.3786     0.2505   5.503 3.74e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For convergence reasons (after trial and error), we used the `bobyqa` optimizer. We do not care so much about the technical details behind the model fitting procedure. Briefly, the model is not fitted using the least squares method anymore, but the maximum likelihood method (see also Westfall p.43-52), which is another way of finding model parameters in parametric statistics. Interestingly, the way to fit the model in front of us is outlined in the help file of the command `glmer` (see `?glmer`).

The syntax `(1 | tank)` means that we fit a random intercept for each tank. For details on the random effect structure definition see p.7 here. You will always see such a term if you model clustered data in the Frequentist setting (at least in R).

AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) are two measures of model fit which both consider the model fit and the number of parameters simultaneously. Both punish the number of parameters in the model, but not equally strong.

In the summary output, we see the values for AIC, BIC and the log-likelihood. The latter is used to fit the model. `-2logLik` is the so-called *deviance* of the model and has - under H_0 - a χ^2 distribution that can be exploited for hypothesis testing. This term takes the place of the residual sum of squares in least squares regression.

The `Std.Dev.` of the `Random effect tank` is 1.568 which should be somewhat similar to the `sigma` in the Bayesian model (1.614989185), which is the case here. The major difference is that in the Frequentist model, this σ is only learned from the data, while in the Bayesian model we have priors too.

The `Fixed effect (Intercept)` (1.3786) is the `logit` of the mean survival probability across all tanks and analog to the `a_bar` (1.3420237045) in the Bayesian model.

`df.resid`is 46, sample size is 48. Two degrees of freedom are lost due to the estimation of the overall `(Intercept)` and the random effect (the standard deviation σ of `tank`).

And of course, at the bottom of the summary output, for the *stargazers* among us, are the `Signif. codes`.

In summary, both (Frequentist and Bayesian) models deliver rather similar results.

We leave it as an exercise (4) to compare the model-estimated survival probabilities in the tanks for both models.

Section 13.2. in the book shows that partial pooling also yields (on average) better predictions compared to no pooling, it's a tradeoff between over- and underfitting (i.e. the raw probabilities in each tank without knowledge of the other tanks and one overall survival probability that ignores differences between tanks completely).

Example: Mulligan manual therapy - paper

We want to replicate the main result of "Mulligan manual therapy added to exercise improves headache frequency, intensity and disability more than exercise alone in people with cervicogenic headache: a randomised trial".

This randomized clinical trial evaluated the effectiveness of adding Mulligan manual therapy to exercise in patients with cervicogenic headache. The study found that combining Mulligan therapy with exercise led to greater improvements in headache frequency, intensity, and disability compared to exercise alone or a control group.

Some comments:

- They used SPSS for data analysis. Unfortunately, no code was provided or another convenient way replicate the results.
- Data *was* provided as Excel file in the appendix.
- At first glance, I could not find a method for sample size calculation in the referenced literature. Let's research this more as an exercise.
- The primary outcome is *headache frequency* (headache days per month). Ad hoc, such a count variable is not very likely to be normally distributed. In the methods section they state "For outcomes with normally distributed data, means and standard deviations were calculated". Table 2 in the paper implies that the primary outcome should be normally distributed. Verify this as exercise using the `shapiro.test` function in R. We (only) do this, since they as well used a version of this test. The primary outcome variable has only 6 unique values ($4-9$) at week 0 and can therefore (strictly speaking) not be normally distributed (normal distribution takes infinitely many values).
- I am not sure what the authors mean exactly with the term "general linear model". Here (p.102) a general linear model is just a multiple linear regression model. My guess is that they did a two-way repeated measures ANOVA.
- There seems to be no checking of the model assumptions in the paper.
- There seems to be no description of handling missing data.

Frequentist attempt to replicate the results. We could try to use this as guideline if we wanted to replicate the two-way repeated measures ANOVA.

Maybe we try to use the `lme4` package first to fit a linear mixed model (LMM), which is the more modern approach.

The model equation should be:

$$\begin{aligned} \text{headache frequency}_{ij} &\sim \text{Normal}(\mu_{ij}, \sigma) \\ \mu_{ij} &= \alpha + \beta_1 \text{group}_i + \beta_2 \text{time}_j + \beta_3 (\text{group}_i \times \text{time}_j) + u_i + \varepsilon_{ij} \end{aligned}$$

- We model the headache frequency of patient i at time point j as normally distributed. It remains to bee seen if a normally distributed headache frequency is a good assumption. We will check this later. A priori, I would be doubtful.

- α is the overall intercept (mean headache frequency at week 0 for the reference group)
- $i = 1, \dots, 99$ are the 99 participants (respectively 91 with complete observations)
- j is the time point (week 0, 4, 13, 26)
- u_i is the random effect for participant i (intercept), which comes from a normal distribution of course
- ε_{ij} is the residual error term, which is also normally distributed. This is a measure of how much the model does not explain.
- The interaction term allows for different intercepts for all combinations of group and time. Note that both `group` and `time` are factors in the model. There are 12 combinations of group and time (3 groups times 4 time points). Be careful with the interpretation of the interaction term. As soon as an interaction term is present, we cannot directly interpret the main effects anymore.

Let's first bring our data in shape (long format for `lmer`) and (for simplicity) kill all rows with missing values (8 observations). Ideally, we would need to argue how we handle missing data!

```
library(readxl)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages -----------------------------------------------------------------
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
```

```
url <- "https://raw.githubusercontent.com/jdegenfellner/Script_QM3_ZHAW/main/data/Chapter_Further_Regres
temp_file <- tempfile(fileext = ".xls")
download.file(url, destfile = temp_file, mode = "wb")
df <- suppressMessages(
  suppressWarnings(
    readxl::read_xls(temp_file, sheet = 2)
  )
)

df <- df[3:dim(df)[1], 1:6]
head(df)
```

```
## # A tibble: 6 x 6
##   `Subject no.` Group `Headache frequency` ...4  ...5  ...6
##           <dbl> <chr> <chr>                <chr> <chr> <chr>
## 1             1 Ex    5                    4     5     4
## 2             2 Ex    7                    6     6     5
## 3             3 Ex    6                    4     7     6
## 4             4 Ex    7                    5     6     5
## 5             5 Ex    7                    7     5     <NA>
## 6             6 Ex    4                    4     4     4
```

```
dim(df)
```

```
## [1] 99  6
```

```r
colnames(df) <- c("ID", "Group", "Week_0",
                  "Week_4", "Week_13", "Week_26")
df
```

```
## # A tibble: 99 x 6
##       ID Group Week_0 Week_4 Week_13 Week_26
##    <dbl> <chr> <chr>  <chr>  <chr>   <chr>
##  1     1 Ex    5      4      5       4
##  2     2 Ex    7      6      6       5
##  3     3 Ex    6      4      7       6
##  4     4 Ex    7      5      6       5
##  5     5 Ex    7      7      5       <NA>
##  6     6 Ex    4      4      4       4
##  7     7 Ex    6      5      5       6
##  8     8 Ex    7      4      4       4
##  9     9 Ex    6      7      7       5
## 10    10 Ex    6      7      6       5
## # i 89 more rows
```

```r
# Omit missing values since they will be thrown out anyways when using lmer!
df <- na.omit(df) # 8 rows (patients) are out!

df_long <- df %>%
  pivot_longer(
    cols = starts_with("Week_"),
    names_to = "time",
    values_to = "headache_frequency"
  ) %>%
  mutate(
    headache_frequency = as.numeric(headache_frequency),
    time = dplyr::recode(time,
                  "Week_0" = 0,
                  "Week_4" = 4,
                  "Week_13" = 13,
                  "Week_26" = 26)
  )

df_long$ID <- as.factor(df_long$ID)
df_long$time_f <- factor(df_long$time, levels = c(0, 4, 13, 26))
df_long
```

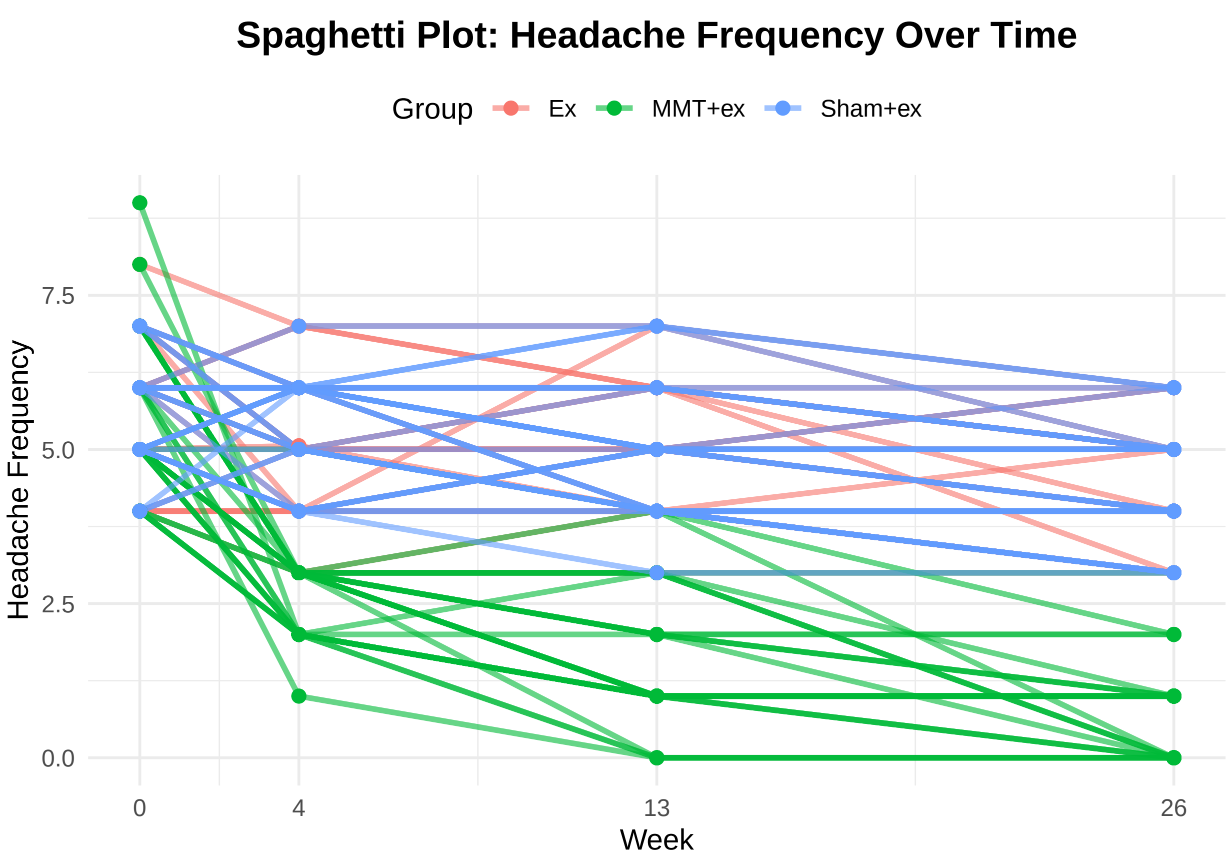
```
## # A tibble: 364 x 5
##    ID    Group  time headache_frequency time_f
##    <fct> <chr> <dbl>              <dbl> <fct>
##  1 1     Ex        0                  5 0
##  2 1     Ex        4                  4 4
##  3 1     Ex       13                  5 13
##  4 1     Ex       26                  4 26
##  5 2     Ex        0                  7 0
##  6 2     Ex        4                  6 4
##  7 2     Ex       13                  6 13
##  8 2     Ex       26                  5 26
##  9 3     Ex        0                  6 0
```

```
## 10 3     Ex        4                   4 4
## # i 354 more rows
```

This seems to be okay. Note that we have 364 lines because of the long format: 91 participants (with complete observations) times 4 time points $(0, 4, 13, 26$ weeks).

Next, we look at the raw data with a spaghetti plot. **It is alway a good idea to know your raw data.** Below, we have plotted the headache frequency colored by group. Time (week) is an integer variable and therefore the distances between the time points are not equal.

```r
ggplot(df_long, aes(x = time, y = headache_frequency, group = ID, color = Group)) +
  geom_line(alpha = 0.6, linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = c(0, 4, 13, 26)) +
  labs(
    title = "Spaghetti Plot: Headache Frequency Over Time",
    x = "Week",
    y = "Headache Frequency",
    color = "Group"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    legend.position = "top"
  ) +
  theme(plot.title = element_text(hjust = 0.5))
```



It seems that the 3 different groups have different slopes which is indicative of an interaction effect between group and time. This means, treatments may differ in their effectiveness. *MMT + ex* seems to be clearly better compared to the other two groups (*Ex* and *Sham + ex*). Since participants were randomly assigned to the groups, this is indicative of a treatment effect.

With respect to our **clustering**, in our long data set, we have repeated measures for each participant (ID) at different time points. There are no repeated treatments though, since each participant is only in one group (of the three) and they do not crossover to one of the other treatments during the trial. *Why do we not need a random intercept for the group?* We are not drawing the groups from a larger population, the groups are fixed and the only ones of interest. The participants on the other hand are drawn from a larger population. Hence, we only consider the random intercept for each participant.

Let's fit the model using the `lme4` package:

```
library(lme4)
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
model <- lmer(headache_frequency ~ Group * time_f + (1 | ID),
              data = df_long)
summary(model) # ref for time: week 0
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: headache_frequency ~ Group * time_f + (1 | ID)
##    Data: df_long
##
## REML criterion at convergence: 993.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0566 -0.5788 -0.0496  0.5513  3.7147
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  ID       (Intercept) 0.3449   0.5872
##  Residual             0.6619   0.8136
## Number of obs: 364, groups:  ID, 91
##
## Fixed effects:
##                        Estimate Std. Error t value
## (Intercept)             5.63333    0.18319  30.751
## GroupMMT+ex            -0.06667    0.25907  -0.257
## GroupSham+ex           -0.05269    0.25697  -0.205
## time_f4                -0.63133    0.21007  -3.005
## time_f13               -0.63333    0.21007  -3.015
## time_f26               -1.26667    0.21007  -6.030
## GroupMMT+ex:time_f4    -2.30200    0.29708  -7.749
## GroupSham+ex:time_f4    0.37327    0.29467   1.267
## GroupMMT+ex:time_f13   -3.33333    0.29708 -11.220
## GroupSham+ex:time_f13  -0.01183    0.29467  -0.040
## GroupMMT+ex:time_f26   -3.53333    0.29708 -11.894
```

```
## GroupSham+ex:time_f26  0.07312    0.29467   0.248
##
## Correlation of Fixed Effects:
##             (Intr) GrMMT+ GrpSh+ tim_f4 tm_f13 tm_f26 GMMT+:_4 GS+:_4 GMMT+:_1
## GroupMMT+ex -0.707
## GroupSham+x -0.713  0.504
## time_f4     -0.573  0.405  0.409
## time_f13    -0.573  0.405  0.409  0.500
## time_f26    -0.573  0.405  0.409  0.500  0.500
## GrpMMT+x:_4  0.405 -0.573 -0.289 -0.707 -0.354 -0.354
## GrpShm+x:_4  0.409 -0.289 -0.573 -0.713 -0.356 -0.356  0.504
## GrpMMT+:_13  0.405 -0.573 -0.289 -0.354 -0.707 -0.354  0.500    0.252
## GrpShm+:_13  0.409 -0.289 -0.573 -0.356 -0.713 -0.356  0.252    0.500  0.504
## GrpMMT+:_26  0.405 -0.573 -0.289 -0.354 -0.354 -0.707  0.500    0.252  0.500
## GrpShm+:_26  0.409 -0.289 -0.573 -0.356 -0.356 -0.713  0.252    0.500  0.252
##             GS+:_1 GMMT+:_2
## GroupMMT+ex
## GroupSham+x
## time_f4
## time_f13
## time_f26
## GrpMMT+x:_4
## GrpShm+x:_4
## GrpMMT+:_13
## GrpShm+:_13
## GrpMMT+:_26  0.252
## GrpShm+:_26  0.500  0.504
```

We are estimating headache frequency while considering repeated measures within patients.

In the random effects part of the output, we see again the standard deviation of the random intercept for each patient (0.5872), which is a measure of the variability of headache frequency between patients. Residual variance (0.6619) is high compared to this.

The so-called `Fixed effects` give the overall intercept in all our factor combinations: `(Intercept)` is the overall mean headache frequency at week 0 (reference week) and Group `Ex` (reference group).

Important for us are the so-called **contrasts** (below). These give us the **model-estimated differences** in (expected) headache frequency between the groups at each time point. To be exact, we are actually interested in the **difference in differences** (DiD) (see Table 3 in the paper):

$\text{headache frequency}_{\text{MMT+ex}} - \text{headache frequency}_{\text{Sham+ex}}$ at week 4**minus**$\text{headache frequency}_{\text{MMT+ex}} - \text{headache frequency}_{\text{Sh}}$

```r
library(emmeans)
emm <- emmeans(model, ~ Group * time_f)
emm_df <- as.data.frame(emm)
emm_df
```

```
##  Group   time_f   emmean        SE     df lower.CL upper.CL
##  Ex      0      5.633333 0.1831917 260.36 5.272607 5.994059
##  MMT+ex  0      5.566667 0.1831917 260.36 5.205941 5.927393
##  Sham+ex 0      5.580645 0.1802128 260.36 5.225785 5.935505
##  Ex      4      5.002000 0.1831917 260.36 4.641274 5.362726
##  MMT+ex  4      2.633333 0.1831917 260.36 2.272607 2.994059
##  Sham+ex 4      5.322581 0.1802128 260.36 4.967721 5.677441
```

```
##   Ex      13     5.000000 0.1831917 260.36 4.639274 5.360726
##   MMT+ex  13     1.600000 0.1831917 260.36 1.239274 1.960726
##   Sham+ex 13     4.935484 0.1802128 260.36 4.580624 5.290344
##   Ex      26     4.366667 0.1831917 260.36 4.005941 4.727393
##   MMT+ex  26     0.766667 0.1831917 260.36 0.405941 1.127393
##   Sham+ex 26     4.387097 0.1802128 260.36 4.032237 4.741957
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
contrast_vec <- list(
  "MMTvsSham_W4_minus_W0" = c(
    0,   # Ex @ 0
    +1,  # MMT+ex @ 0
    -1,  # Sham+ex @ 0
    0,   # Ex @ 4
    -1,  # MMT+ex @ 4
    +1,  # Sham+ex @ 4
    0, 0, 0, 0, 0, 0 # ignore reset (13 & 26)
  )
)

contrast(emm, method = contrast_vec, infer = c(TRUE, FALSE))
```

```
##  contrast              estimate    SE  df lower.CL upper.CL
##  MMTvsSham_W4_minus_W0     2.68 0.295 264      2.1     3.26
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

Bonferroni correction is used to adjust the p-values for multiple comparisons (one just divides the "significance"-level by the number of comparisons).

We now look at Table 3 in the paper, first row, first column. There we see the mean between-group difference in headache frequency at week 4 compared to week 0 and the two groups *MMT + ex* and *MMT Sham + ex* are compared.

- The mean difference (in the paper) is -3 (95% CI: -3 to -2) days/month.
- In our `contrast` output, we find the row `MMTvsSham_W4_minus_W0     2.68 0.295 264      2.1 3.26`.

After rounding to integers we get the same results.

If you want to see the **random effects** of all 91 participants explicitely:

```
ranef(model)  # random effects
```

```
## $ID
##      (Intercept)
## 1  -0.3382077044
## 2   0.6754017992
## 3   0.5064668819
## 4   0.5064668819
```

```
## 6  -0.6760775389
## 7   0.3375319647
## 8  -0.1692727871
## 9   0.8443367165
## 10  0.6754017992
## 12  0.5064668819
## 13 -0.0003378698
## 14  1.0132716337
## 15  0.1685970474
## 16 -1.1828822907
## 17 -0.5071426216
## 18  0.3375319647
## 19 -0.1692727871
## 20 -0.6760775389
## 21 -0.1692727871
## 22 -0.0003378698
## 23 -0.6760775389
## 24 -0.6760775389
## 25 -0.8450124561
## 26  0.3375319647
## 28  0.6754017992
## 29  0.3375319647
## 30 -0.1692727871
## 31 -0.3382077044
## 32 -0.0003378698
## 33 -0.3280716093
## 34 -0.6025345382
## 35  0.5800098826
## 36  0.5800098826
## 37  0.0732051308
## 39  0.4110749653
## 40  0.4110749653
## 41  0.2421400481
## 42  0.0732051308
## 43 -0.0957297864
## 44  0.4110749653
## 46 -0.2646647037
## 47 -0.6025345382
## 48  0.5800098826
## 49  0.2421400481
## 50 -0.2646647037
## 51  0.2421400481
## 52 -0.2646647037
## 53 -0.6025345382
## 54  0.0732051308
## 55 -0.0957297864
## 56 -0.2646647037
## 57 -0.6025345382
## 58  0.9178797171
## 60 -0.0957297864
## 61  0.0732051308
## 62 -0.2646647037
## 63 -0.4335996210
## 64  0.2421400481
```

```
## 65 -0.2646647037
## 66 -0.4335996210
## 67 -0.2070815115
## 68 -0.2070815115
## 69 -0.3760164287
## 70 -0.2070815115
## 71  0.2997232403
## 72  0.2997232403
## 73 -0.3760164287
## 74  0.4686581576
## 75 -0.2070815115
## 76 -0.2070815115
## 77  0.4686581576
## 78  0.9754629093
## 79  0.4686581576
## 81 -0.0381465942
## 82  0.2997232403
## 83 -0.0381465942
## 84 -0.3760164287
## 85 -0.0381465942
## 86 -0.5449513460
## 87 -0.0381465942
## 88  0.4686581576
## 90 -0.2070815115
## 91  0.9754629093
## 92 -0.2070815115
## 93  0.4686581576
## 94 -0.5449513460
## 95  0.8065279921
## 96 -0.7138862633
## 97 -0.2070815115
## 98 -0.7138862633
## 99 -0.5449513460
##
## with conditional variances for "ID"
```

```r
re <- ranef(model)
str(re$ID) # 91 obs, 91 participants
```

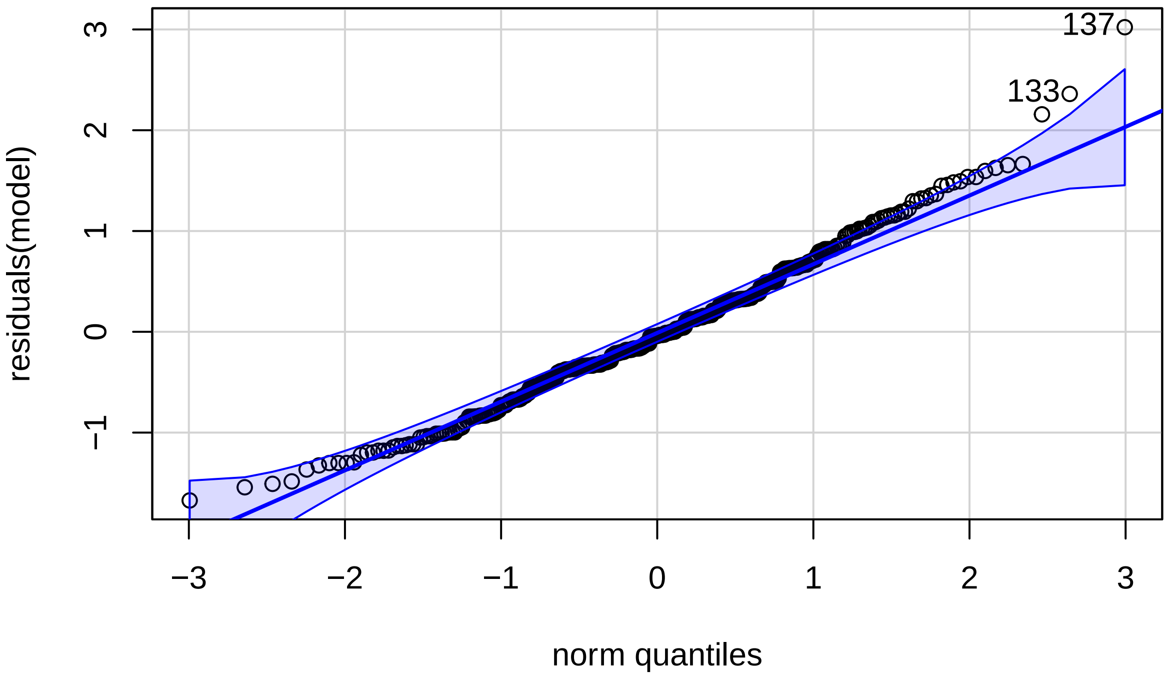
```
## 'data.frame':    91 obs. of  1 variable:
##  $ (Intercept): num  -0.338 0.675 0.506 0.506 -0.676 ...
##   - attr(*, "postVar")= num [1, 1, 1:91] 0.112 0.112 0.112 0.112 0.112 ...
```

Posterior Predictive checks and **residuals** look surprisingly good:

```r
library(car)
```

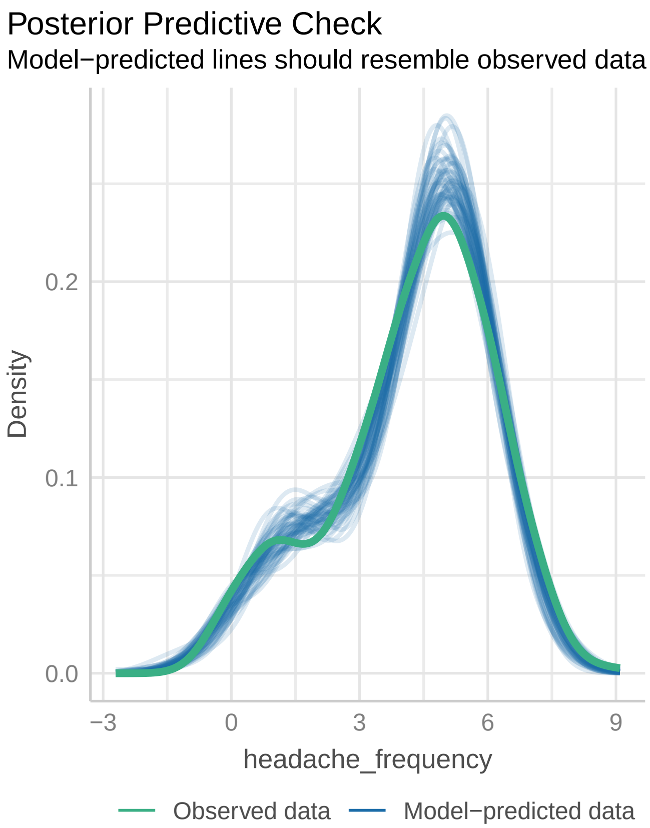
```
## Loading required package: carData
```

```r
library(performance)
qqPlot(residuals(model))
```




```
## [1] 137 133
```

```
check_model(model, check = "pp_check")
```



The density estimators might be a bit misleading since we only have a rather small number of different levels and of course a count variable can not be negative.

Bayesian attempt to replicate the results We can also try to fit the model using the `rethinking` package.

Model equations and priors:

$$
\begin{aligned}
\text{headache frequency}_{ij} &\sim \text{Normal}(\mu_{ij}, \sigma) \\
\mu_{ij} &= \alpha + \beta_1 \text{group}_i + \beta_2 \text{time}_j + \beta_3 (\text{group}_i \times \text{time}_j) + u_i \\
\alpha &\sim \text{Normal}(5, 1.5) \\
\beta_1[group] &\sim \text{Normal}(0, 1.5) \\
\beta_2[time] &\sim \text{Normal}(0, 1.5) \\
\beta_3[group * time] &\sim \text{Normal}(0, 1.5) \\
u_i &\sim \text{Normal}(0, \sigma_u) \\
\sigma_u &\sim \text{Exponential}(1) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

The priors for β_3 is hard since there is no natural meaning to it. Note that we could also combine α and the u_i as we did before in the partial pooling case for reliability. This should deliver similar results (exercise 10). With this model we now fit intercepts for every group (3), every time point (4) and for all $3 * 4 = 12$ interaction combinations. Since we are not working with a reference level for the factors (as in `lmer`), you see all $3, 4$ and 12 (mean posterior) levels of the intercepts in the output of `precis(m)`. Refer to section 5.3 (Categorical variables) in the book *Rethinking* for more details.

```r
library(rethinking)

df_long <- df_long %>%
  dplyr::group_by(Group, time) %>%
  dplyr::mutate(interaction = cur_group_id()) %>%
  dplyr::ungroup()
# cur_group_id() gives a unique numeric identifier for the current group.

dat <- list(
  H = df_long$headache_frequency,
  group = as.integer(as.factor(df_long$Group)),   # 1 = Ex, 2 = MMT+ex, 3 = Sham+ex
  time = as.integer(df_long$time_f),              # 1 = Woche 0, ..., 4 = Woche 26
  ID = as.integer(as.factor(df_long$ID)),
  N = nrow(df_long),
  N_ID = length(unique(df_long$ID)),
  N_group = length(unique(df_long$Group)),
  N_time = length(unique(df_long$time_f)),
  interaction = df_long$interaction
)

m <- ulam(
  alist(
    H ~ normal(mu, sigma),
    mu <- alpha +
      beta_group[group] + beta_time[time] + beta_interaction[interaction] + u[ID],
    # random intercept for each participant
    u[ID] ~ normal(0, sigma_ID),
    # Priors
    alpha ~ normal(0, 10),
    beta_group[group] ~ normal(0, 2),
    beta_time[time] ~ normal(0, 2),
    beta_interaction[interaction] ~ normal(0, 2), # difficult
```

```r
    sigma ~ exponential(1),
    sigma_ID ~ exponential(1)
  ),
  data = dat,
  chains = 4,
  cores = 4
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:   1 / 1000 [  0%]  (Warmup)

## Chain 1 Informational Message: The current Metropolis proposal is about to be rejected because of th

## Chain 1 Exception: normal_lpdf: Scale parameter is 0, but must be positive! (in '/var/folders/pm/jd6

## Chain 1 If this warning occurs sporadically, such as for highly constrained variable types like cova

## Chain 1 but if this warning occurs often then your model may be either severely ill-conditioned or m

## Chain 1

## Chain 2 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 3 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 4 Iteration:   1 / 1000 [  0%]  (Warmup)
## Chain 1 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 2 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 3 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 4 Iteration: 100 / 1000 [ 10%]  (Warmup)
## Chain 2 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 4 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 3 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 1 Iteration: 200 / 1000 [ 20%]  (Warmup)
## Chain 2 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 3 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 4 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 1 Iteration: 300 / 1000 [ 30%]  (Warmup)
## Chain 2 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 3 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 4 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 1 Iteration: 400 / 1000 [ 40%]  (Warmup)
## Chain 2 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 2 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 3 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 3 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 4 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 4 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 1 Iteration: 500 / 1000 [ 50%]  (Warmup)
## Chain 1 Iteration: 501 / 1000 [ 50%]  (Sampling)
## Chain 2 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 3 Iteration: 600 / 1000 [ 60%]  (Sampling)
```

```
## Chain 4 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 1 Iteration: 600 / 1000 [ 60%]  (Sampling)
## Chain 2 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 3 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 4 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 1 Iteration: 700 / 1000 [ 70%]  (Sampling)
## Chain 2 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 3 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 4 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 1 Iteration: 800 / 1000 [ 80%]  (Sampling)
## Chain 2 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 3 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 4 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 1 Iteration: 900 / 1000 [ 90%]  (Sampling)
## Chain 2 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 2 finished in 4.0 seconds.
## Chain 3 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 4 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 3 finished in 4.0 seconds.
## Chain 4 finished in 4.1 seconds.
## Chain 1 Iteration: 1000 / 1000 [100%]  (Sampling)
## Chain 1 finished in 4.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.2 seconds.
## Total execution time: 4.4 seconds.
```

```
precis(m, depth = 2)
```

```
##                          mean         sd        5.5%        94.5%      rhat
## u[1]             -0.337127662 0.34253797 -0.89186510  0.196422865 1.0035274
## u[2]              0.674342075 0.35125639  0.11962043  1.244117500 1.0031747
## u[3]              0.491657427 0.34898385 -0.06927222  1.059163300 1.0052700
## u[4]              0.497952044 0.33483891 -0.03150310  1.027160500 1.0009816
## u[5]             -0.670080220 0.35976361 -1.24808295 -0.097855198 1.0070188
## u[6]              0.329085035 0.34407160 -0.21335876  0.873768540 1.0017772
## u[7]             -0.170025941 0.35656416 -0.74680623  0.386595810 1.0052362
## u[8]              0.838927993 0.34992731  0.29131198  1.402978700 1.0030951
## u[9]              0.670779191 0.35925771  0.11371105  1.235624300 0.9998655
## u[10]             0.496734418 0.34734841 -0.05229111  1.044467650 1.0010689
## u[11]            -0.006990641 0.34314778 -0.54261154  0.554184980 1.0012468
## u[12]             1.003282898 0.35805482  0.43562923  1.591864200 1.0000800
## u[13]             0.169631558 0.33674672 -0.38062077  0.700714635 1.0082336
## u[14]            -1.186208388 0.36995610 -1.79068940 -0.585241545 1.0012515
## u[15]            -0.512298440 0.34365921 -1.03975035  0.039454993 1.0041191
## u[16]             0.332723045 0.33549826 -0.21504814  0.860813935 1.0003011
## u[17]            -0.174759220 0.36312307 -0.75137590  0.402268330 0.9989290
## u[18]            -0.676208426 0.35967626 -1.26383760 -0.095767189 1.0059114
## u[19]            -0.170374298 0.33194972 -0.70326458  0.369975015 1.0092759
## u[20]            -0.003196991 0.35575212 -0.56601115  0.559372810 1.0017752
## u[21]            -0.679477365 0.34249532 -1.22262290 -0.130318355 1.0007932
## u[22]            -0.673394355 0.34111803 -1.22198145 -0.132575645 1.0008262
## u[23]            -0.848459927 0.35540468 -1.40995260 -0.303205295 1.0022640
## u[24]             0.339492360 0.35522959 -0.21720090  0.921766215 1.0035994
```

```
## u[25]                    0.659402833 0.35378237  0.08631138  1.196533900 1.0030242
## u[26]                    0.340822754 0.34130561 -0.20853252  0.880764200 1.0068952
## u[27]                   -0.179408642 0.35703297 -0.74347031  0.391867855 1.0023312
## u[28]                   -0.342911514 0.34164149 -0.88958953  0.211778335 1.0028246
## u[29]                   -0.003645727 0.35646968 -0.56116811  0.598281270 0.9990745
## u[30]                   -0.329120594 0.37162675 -0.93867122  0.252788045 0.9992046
## u[31]                   -0.606169535 0.35887932 -1.17946550 -0.034321859 1.0022508
## u[32]                    0.570678899 0.35406015  0.01885522  1.149743700 1.0019715
## u[33]                    0.577722808 0.33788489  0.04612221  1.126430600 1.0008750
## u[34]                    0.067801066 0.33541390 -0.48123591  0.598514165 0.9992811
## u[35]                    0.407549683 0.33577145 -0.12118522  0.945824120 1.0072372
## u[36]                    0.408973984 0.35574410 -0.13742707  0.958021505 0.9997201
## u[37]                    0.233988305 0.34328821 -0.30284449  0.787055150 1.0043536
## u[38]                    0.069778657 0.33824633 -0.47036092  0.626452770 1.0009169
## u[39]                   -0.104982899 0.34687704 -0.65537833  0.465851015 0.9991588
## u[40]                    0.401220184 0.35403523 -0.16590124  0.962760120 1.0021123
## u[41]                   -0.267422615 0.34554958 -0.81570580  0.270627900 1.0048218
## u[42]                   -0.601348301 0.36115498 -1.19242190 -0.022099343 1.0012401
## u[43]                    0.578815033 0.35573172  0.02981857  1.128717050 1.0064939
## u[44]                    0.230252398 0.33575334 -0.29262760  0.752231225 1.0005750
## u[45]                   -0.263098507 0.35819708 -0.84565283  0.295185445 0.9991398
## u[46]                    0.234821740 0.34872834 -0.31475964  0.790490705 1.0028769
## u[47]                   -0.268265025 0.34821049 -0.82714671  0.279794470 1.0021875
## u[48]                   -0.597613448 0.35751500 -1.16591385 -0.008009208 0.9991286
## u[49]                    0.080071408 0.35514759 -0.47879174  0.656419440 1.0031102
## u[50]                   -0.094905409 0.33672574 -0.60775762  0.452160590 1.0029882
## u[51]                   -0.266431370 0.33375453 -0.81188447  0.258864415 1.0013264
## u[52]                   -0.605389263 0.34233355 -1.15478245 -0.052617640 1.0002843
## u[53]                    0.913967093 0.35936945  0.36223567  1.503156650 1.0049616
## u[54]                   -0.094259729 0.34229201 -0.64496014  0.459361660 1.0024303
## u[55]                    0.071237087 0.34062302 -0.48547989  0.602445810 1.0025961
## u[56]                   -0.265471686 0.33791738 -0.81450508  0.275555105 1.0027965
## u[57]                   -0.437171439 0.34171970 -0.99169718  0.106894200 1.0020578
## u[58]                    0.235589746 0.34373466 -0.30753506  0.779752095 1.0025677
## u[59]                   -0.264957986 0.34166227 -0.80043977  0.289044265 1.0002387
## u[60]                   -0.434860207 0.35231561 -1.02551070  0.125031020 1.0006584
## u[61]                   -0.211657616 0.35694504 -0.78402871  0.356888035 1.0011604
## u[62]                   -0.204951313 0.33923523 -0.74204415  0.348087130 1.0013705
## u[63]                   -0.369092767 0.33889663 -0.90803666  0.179362200 0.9995762
## u[64]                   -0.210113713 0.35854192 -0.78325608  0.339609505 1.0016976
## u[65]                    0.297584435 0.34160535 -0.24819650  0.820526535 1.0054150
## u[66]                    0.293148426 0.34625405 -0.26275153  0.856682950 1.0022401
## u[67]                   -0.368874502 0.35930948 -0.94006124  0.207946960 1.0039553
## u[68]                    0.467242119 0.33855613 -0.08058247  0.995832900 1.0001017
## u[69]                   -0.213195545 0.35650653 -0.77835188  0.360084900 1.0068283
## u[70]                   -0.204576019 0.33595236 -0.73389874  0.330641950 1.0027028
## u[71]                    0.458132769 0.35804702 -0.09809696  1.026747400 1.0015276
## u[72]                    0.967604120 0.35150263  0.41228810  1.544895350 1.0036365
## u[73]                    0.467311564 0.35026389 -0.11485745  1.025121150 1.0008388
## u[74]                   -0.033076253 0.35060544 -0.59562588  0.507460800 1.0043722
## u[75]                    0.303074556 0.35613252 -0.26112126  0.880553325 1.0054639
## u[76]                   -0.036929163 0.35367922 -0.58573388  0.511903855 1.0029966
## u[77]                   -0.373287324 0.35840503 -0.95037028  0.193829730 1.0010101
## u[78]                   -0.032458450 0.34698370 -0.58446753  0.521201860 1.0040202
```

```
## u[79]                -0.546493613 0.36181763 -1.14313820  0.034444768 1.0001499
## u[80]                -0.038379977 0.34995873 -0.58714573  0.503523295 1.0024587
## u[81]                 0.467896545 0.34079246 -0.08699699  1.016716800 1.0042446
## u[82]                -0.211755342 0.34951586 -0.75844506  0.343520360 1.0085883
## u[83]                 0.974388486 0.35764266  0.40732480  1.548196150 1.0056134
## u[84]                -0.211505140 0.33609252 -0.73853546  0.315226090 1.0003503
## u[85]                 0.466157139 0.33864727 -0.07700219  1.006821550 1.0001420
## u[86]                -0.543142722 0.35006748 -1.08433140 -0.001150085 1.0038356
## u[87]                 0.805065455 0.35506798  0.26339479  1.378855400 1.0007497
## u[88]                -0.720224151 0.34494004 -1.28103730 -0.174145445 1.0024170
## u[89]                -0.202983321 0.35161723 -0.77552749  0.363461965 1.0096548
## u[90]                -0.705823656 0.36545582 -1.30790325 -0.117473225 1.0003789
## u[91]                -0.544537274 0.35447761 -1.09764655  0.016453738 1.0010408
## alpha                 4.073470673 1.63533357  1.39442940  6.621043850 1.0041581
## beta_group[1]         0.700958560 1.36525094 -1.44879995  2.819580400 1.0046629
## beta_group[2]        -1.223744607 1.40289182 -3.44550770  1.019537950 1.0030525
## beta_group[3]         0.738168905 1.38288555 -1.50685520  2.926935750 1.0008938
## beta_time[1]          1.086408581 1.30035282 -0.93246969  3.185274750 1.0047331
## beta_time[2]          0.114308402 1.31153497 -1.98372220  2.189992650 1.0025640
## beta_time[3]         -0.210912101 1.30929417 -2.30947925  1.873626000 1.0025946
## beta_time[4]         -0.748129048 1.34076457 -2.91102320  1.447518600 0.9994714
## beta_interaction[1]  -0.219857085 1.23184458 -2.18371195  1.729494650 1.0004486
## beta_interaction[2]   0.114858528 1.25016076 -1.87413885  2.085342800 0.9993658
## beta_interaction[3]   0.436410183 1.24523722 -1.53193140  2.430499150 1.0039685
## beta_interaction[4]   0.341537666 1.20704123 -1.54806720  2.275814950 0.9996896
## beta_interaction[5]   1.620060847 1.21874769 -0.34887801  3.548851750 1.0009216
## beta_interaction[6]  -0.325934696 1.31831699 -2.42140110  1.847793300 1.0008627
## beta_interaction[7]  -1.034910661 1.26638950 -2.99352605  1.061350100 1.0018636
## beta_interaction[8]  -1.323765896 1.27747077 -3.36236510  0.738125685 1.0015855
## beta_interaction[9]  -0.318879117 1.16928180 -2.06335695  1.582086250 1.0007553
## beta_interaction[10]  0.392691800 1.26875161 -1.55963840  2.417551300 1.0016131
## beta_interaction[11]  0.336247708 1.24056041 -1.59567965  2.320634000 1.0037643
## beta_interaction[12]  0.321490710 1.24922681 -1.72515280  2.372296550 1.0006902
## sigma                 0.816356153 0.03675947  0.76131098  0.878293245 1.0018290
## sigma_ID              0.591733608 0.06760259  0.48848780  0.701842330 1.0055118
##                      ess_bulk
## u[1]                 4267.176
## u[2]                 3674.808
## u[3]                 4629.296
## u[4]                 4581.993
## u[5]                 4150.488
## u[6]                 3298.611
## u[7]                 5903.838
## u[8]                 3980.020
## u[9]                 3964.515
## u[10]                4157.856
## u[11]                4102.522
## u[12]                3919.616
## u[13]                4533.115
## u[14]                3183.741
## u[15]                3969.164
## u[16]                3754.100
## u[17]                4075.293
## u[18]                4107.628
```

```
## u[19]                    3624.178
## u[20]                    6214.550
## u[21]                    3793.582
## u[22]                    3894.361
## u[23]                    4239.607
## u[24]                    3566.625
## u[25]                    3668.775
## u[26]                    4470.021
## u[27]                    5028.696
## u[28]                    4279.150
## u[29]                    5021.594
## u[30]                    6026.773
## u[31]                    3695.057
## u[32]                    4947.610
## u[33]                    4013.000
## u[34]                    4321.461
## u[35]                    5426.517
## u[36]                    4292.999
## u[37]                    3683.690
## u[38]                    3794.206
## u[39]                    4715.973
## u[40]                    4337.662
## u[41]                    4728.943
## u[42]                    4082.233
## u[43]                    3776.714
## u[44]                    4440.680
## u[45]                    4480.940
## u[46]                    4646.374
## u[47]                    3501.713
## u[48]                    3785.334
## u[49]                    4689.774
## u[50]                    4149.231
## u[51]                    4052.982
## u[52]                    3336.371
## u[53]                    3230.555
## u[54]                    3530.138
## u[55]                    4601.901
## u[56]                    3813.960
## u[57]                    4234.207
## u[58]                    4141.634
## u[59]                    4284.546
## u[60]                    4076.530
## u[61]                    3491.830
## u[62]                    3930.842
## u[63]                    3412.521
## u[64]                    3935.857
## u[65]                    3476.553
## u[66]                    3785.804
## u[67]                    3916.920
## u[68]                    3761.387
## u[69]                    3635.739
## u[70]                    3265.313
## u[71]                    3747.628
## u[72]                    2839.624
```

```
## u[73]                 4194.028
## u[74]                 3507.219
## u[75]                 3310.016
## u[76]                 4802.445
## u[77]                 3884.374
## u[78]                 4081.117
## u[79]                 3519.576
## u[80]                 3616.462
## u[81]                 3925.732
## u[82]                 4348.337
## u[83]                 3635.431
## u[84]                 3506.617
## u[85]                 3968.603
## u[86]                 3300.875
## u[87]                 3349.835
## u[88]                 3105.302
## u[89]                 3508.942
## u[90]                 4115.687
## u[91]                 3429.681
## alpha                 1217.547
## beta_group[1]         1455.798
## beta_group[2]         1785.893
## beta_group[3]         1615.554
## beta_time[1]          1616.171
## beta_time[2]          1681.931
## beta_time[3]          1744.552
## beta_time[4]          1609.315
## beta_interaction[1]   1917.639
## beta_interaction[2]   1956.421
## beta_interaction[3]   1637.661
## beta_interaction[4]   1688.850
## beta_interaction[5]   1852.717
## beta_interaction[6]   1879.072
## beta_interaction[7]   1799.741
## beta_interaction[8]   1916.364
## beta_interaction[9]   1872.211
## beta_interaction[10] 1939.257
## beta_interaction[11] 1594.497
## beta_interaction[12] 1770.925
## sigma                 2361.132
## sigma_ID              1199.424
```

`precis(m, depth = 2)` confirms that `sigma`and `sigma_ID` are in line with `lmer`.

Next, we want to know the posterior distribution (respectively the mean and credible interval) of the difference in difference:

$$(MMT + ex - Sham + ex)_{\text{week}_4} - (MMT + ex - Sham + ex)_{\text{week}_0}$$

The result is rather similar to the Frequentist model.

```r
# find correct indices for the interaction terms:

levels(as.factor(df_long$Group)) # 2 minus 3
```

```
## [1] "Ex"      "MMT+ex"  "Sham+ex"
```

```r
intdf <- df_long %>%
  dplyr::group_by(Group, time) %>%
  dplyr::select(Group, time) %>%
  unique()
intdf$interaction_ID <- 1:nrow(intdf)

# Extract posterior samples
post <- extract.samples(m)

# -------------------------------
# Compute group differences at week 4 and week 0
# -------------------------------

# MMT+ex vs Sham+ex at week 4:
# Index 6 = MMT+ex, week 4
# Index 10 = Sham+ex, week 4
diff_w4 <- post$beta_group[,2] - post$beta_group[,3] +
           post$beta_interaction[,6] - post$beta_interaction[,10]

# MMT+ex vs Sham+ex at week 0:
# Index 5 = MMT+ex, week 0
# Index 9 = Sham+ex, week 0
diff_w0 <- post$beta_group[,2] - post$beta_group[,3] +
           post$beta_interaction[,5] - post$beta_interaction[,9]

# -------------------------------
# Difference-in-Differences
# -------------------------------

diff_in_diff <- diff_w4 - diff_w0

# Summarize posterior distribution (mean and 95% credible interval)
precis(data.frame(diff_in_diff), prob = 0.95)
```

```
##                   mean        sd      2.5%     97.5%   histogram
## diff_in_diff -2.657566 0.2947512 -3.252686 -2.065504
```

Logistic Regression

Simple Logistic Regression

Another basic regression model is the **logistic regression**. There are entire books on the topic.

Watch these videos as an introduction: 1 2.

In our Bayesian (rethinking) framework, this topic is not really new. We have already used it in our tadpole example when we modeled the survival of tadpoles in tanks. There, our outcome variable was the *number of surviving tadpoles* in a tank, which is a count variable and was modeled using a binomial distribution.

In logistic regression, we just model a **binary** outcome variable, with 0 and 1 (*Yes*/*No*; *died*/*survived*...) as possible values.

Of course, the great disadvantage of this model is the low information content of the outcome variable, which can only take two values. We could then ask ourselves, how the probability of a 1 (e.g. survival) changes with respect to the predictors.

Obviously, the classical linear regression makes no sense for this kind of data.

Let's quickly invent an example:

```
set.seed(332)
AGE <- rnorm(100) # standardized age of 100 participants
y <- rbinom(100, size = 1, prob = inv_logit(AGE))
plot(AGE, y, main = "Outcome variable with least squares regression line",
     xlab = "x", ylab = "inv_logit(AGE)", col = "blue", pch = 19)
abline(lm(y ~ AGE), col = "red", lwd = 2)
```

The higher the `AGE`, the higher the probability of a 1 (heart attaack or something). In this case (of a simulation) we know the trajectory of the *true* probabilities varying with AGE. Of course, normally, we do not know this but conceptually asssume there are such true probabilities.

The red line is the **least squares regression** line, which **does not make much sense** here: The outcome can only be 0 or 1 but the regression line would estimate all values in between plus values outside of this range, which is not possible.

We therefore bend our reality a bit and **model the probability of a** 1 ($= p_i = \mathbb{E}(Y_i)$ if Y_i is Bernoulli distributed). Of course, then we have the next problem, if we try to model the probability of a 1 *directly* using the linear regression model ($\mathbb{P}(Y_i = 1) = \beta_0 + \beta_1 x_1 + \cdots \beta_p x_p$). The linear predictor ($\beta_0 + \beta_1 x_1 + \cdots \beta_p x_p$) could easily lie outside of the range $[0, 1]$ (for adequate βs and covariate values).

Therefore, we use the **logit function** to transform the the probability from $[0, 1]$ to $(-\infty, +\infty)$. This transformed range can be conveniently described using a linear predictor again. The left side and the right side of the equation are now equal in range ($(-\infty, +\infty)$). Very nice.

The model is then ($p = 1$ for the simple logistic regression with only one predictor):

$$\begin{aligned}\mathbb{P}(Y_i = 1|x_{i1}, \dots, x_{ip}) &= \mathbb{E}(Y_i|x_{i1}, \dots, x_{ip}) = p_i \\ \text{logit}(p_i) &= \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}\end{aligned}$$

where p_i is the probability of a 1 for observation i and $\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right)$.

Notice that we do not need a random error term ε_i anymore, since these kinds of models (so-called generalized linear models; GLM) are estimated using the maximum likelihood method. See exercise 15.

Frequentist version In current literature, you will see the model estimated mostly in the **Frequentist setting**:

```r
modlog <- glm(y ~ AGE, family = binomial(link = "logit"))
summary(modlog)
```

```
##
## Call:
## glm(formula = y ~ AGE, family = binomial(link = "logit"))
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.2992     0.2352   1.272    0.203
## AGE           1.5215     0.3588   4.240 2.23e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 137.63  on 99  degrees of freedom
## Residual deviance: 108.20  on 98  degrees of freedom
## AIC: 112.2
##
## Number of Fisher Scoring iterations: 4
```

- `glm`stands for *generalized linear model*.
- `family = binomial(link = "logit")` specifies that we want to use the binomial distribution with the `logit` link function g. In generalized linear models, the link function is used to connect the linear predictor (the right side of the equation) to the expected value of the response variable: In our case, the expected value of a Bernoulli-distributed variable (which is binary) is the probability of a 1: p_i.

  In general:

  $$\mathbb{E}(Y|X) = g^{-1}(X\beta)$$

  Think back to exercise 5 in the reliability chapter and go to exercise 11.
- `Coefficients` are the estimated parameters of the model **on the logit scale**.

  Interpretation: Two participants with a difference of 1 standard deviation (since we standardized age) in their age, have an expected difference of $\beta_1 = 1.5215$ in the log-odds of having a heart attack.

  If we want the **odds ratio**, we can just exponentiate:

  $$log(\frac{p_i}{1-p_i}) = \beta_0 + \beta_1 AGE$$

  $$\text{Odds for a } 1 = \frac{p_i}{1-p_i} = e^{\beta_0 + \beta_1 AGE} = e^{\beta_0} \cdot e^{\beta_1 AGE}$$

If you add 1 to AGE:

$$\text{Odds for a 1 with } (AGE+1) = e^{\beta_0} \cdot e^{\beta_1(AGE+1)} = e^{\beta_0} \cdot e^{\beta_1 AGE} \cdot e^{\beta_1} = \text{Odds for a } 1 \cdot e^{\beta_1}$$

So, **on the odds scale**, you **multiply the odds for a** 1 **by** e^{β_1}.

We can also work **directly on the probability scale**: The difference in the probability of a 1 for two participants with a difference of 1 year in their age is:

$$\mathbb{P}(Y_i = 1|AGE+1) - \mathbb{P}(Y_i = 1|AGE) =$$

$$\text{inv\_logit}(\beta_0 + \beta_1(AGE+1)) - \text{inv\_logit}(\beta_0 + \beta_1 AGE)$$

This difference is not constant (see graph below, red line), but depends on the value of AGE, since the inverse logit function is non-linear.

- `Null deviance` is the deviance ($= -2$log-likelihood) of the model with only the intercept (no predictors). The deviance is a measure of how well the model fits the data.

```
-2 * logLik(glm(y ~ 1, family = binomial(link = "logit")))
```

```
## 'log Lik.' 137.6278 (df=1)
```

- `Residual deviance` is the deviance of the model with the predictors.

```
-2 * logLik(modlog)
```

```
## 'log Lik.' 108.1987 (df=2)
```

- `AIC` is the Akaike Information Criterion, which is a measure of the model's goodness of fit, penalized for the number of parameters. Lower AIC values indicate a better fit.

```
-2 * logLik(modlog) + 2 * length(coef(modlog))
```

```
## 'log Lik.' 112.1987 (df=2)
```

The more parameters, the higher the AIC, the worse the model fit. AIC punishes model complexity.

We can calculate the probabilities estimated by data:

```
predicted_probabilities <- predict(modlog, type = "response") # response is the probability p_i here.
plot(AGE, y, main = "Raw data and predicted probabilities",
     xlab = "x", ylab = "inv_logit(AGE)", col = "blue", pch = 19)
points(AGE, predicted_probabilities, col = "red", pch = 19)
```